

The background of the slide is a high-resolution image of Jupiter's atmosphere, showing swirling cloud patterns in shades of brown, tan, and white against a dark blue space background. The image is cropped into a semi-circular shape at the top.

Tutorial 3 Mini-Lecture: Atmospheres

The concepts behind Worksheet 3 • Lectures 5–6

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What Worksheet 3 trains

Four problems, all built on Lectures 5–6. The physics you need, per problem:

1. **How thick is an atmosphere?:** scale height, hydrostatic balance, column mass.
2. **Sunlight in, infrared out:** equilibrium temperature, the greenhouse effect.
3. **From vapour to cloud base:** Clausius-Clapeyron, dew point, the LCL, the saturation-curve figure.
4. **Climate on the edge:** the faint young Sun, ice-albedo bistability, the carbonate-silicate thermostat.

Every problem has the shape and level of one exam question. All parts are analytical; no computer is needed.

How thick is an atmosphere?

Two workhorses of Lecture 5:

$$H = \frac{k_B T}{\mu m_u g} \quad P_0 = m_{\text{col}} g$$

- H compares thermal energy against gravitational confinement: Earth 8.4 km, Titan 21 km.
- Integrating $dP = -\rho g dz$ over the column: surface pressure is the weight of the air overhead.

Watch the ratio, not the terms

$H \propto T/g$: when two effects pull in opposite directions, put numbers on both ratios before deciding which one wins. Problem 1(b) turns exactly on this.

Ten tonnes of air stand on every square metre of Earth; the scale height says how thickly it is spread.

Sunlight in, infrared out

Balance absorbed sunlight against emitted infrared:

$$(1 - A) \frac{S}{4} = \sigma T_{\text{eq}}^4$$

- S falls with the inverse square of distance; A is the Bond albedo.
- Venus: brilliant clouds give $T_{\text{eq}} = 227 \text{ K}$, colder than Earth's 255 K , above a 737 K surface.

The greenhouse in one sentence

The atmosphere passes sunlight but blocks the surface's infrared, so the surface must heat up until its share leaks through.

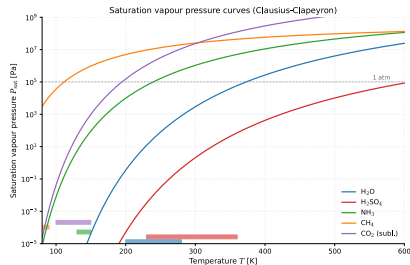
The place to test any claim about the greenhouse budget is the top of the atmosphere: check whether absorbed sunlight and outgoing infrared still balance there.

From vapour to cloud base

The blackboard result of Lecture 6:

$$P_{\text{sat}}(T) = P_{\text{ref}} e^{-\frac{L_v}{R_v} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right)}$$

- Invert it for the dew point T_d ; rising air cools at $\Gamma_d = 9.8 \text{ K km}^{-1}$ until it gets there.
- $z_{\text{LCL}} = (T - T_d)/\Gamma_d$: the cloud base. Problem 3(a) runs this chain from surface humidity to cloud-base height.



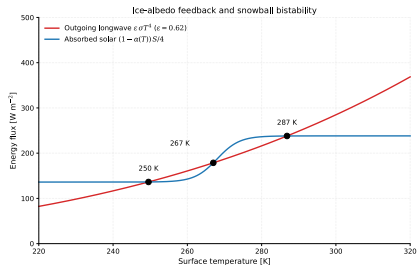
The saturation curves printed on the sheet: Problem 3(b) finds the species whose curve passes the given (T, P) point, the body follows from the given data. Argue the species from the curves.

One exponential governs every cloud in the solar system; L_v/R_v sets how sharply it bites.

Climate on the edge

Long-term climate is a balance with feedbacks:

- Young Sun at 71%: $T_{\text{eq}} = 234 \text{ K}$, yet the rocks record liquid water: the faint young Sun paradox.
- Ice-albedo feedback: two stable climates under the same Sun.
- Carbonate-silicate thermostat: warmer $\rightarrow$ faster weathering $\rightarrow$ less $\text{CO}_2 \rightarrow$ cooler.



The bistability diagram printed on the sheet: Problem 4(b) classifies each crossing from the local slopes, by asking whether absorption or emission grows faster after a small nudge in temperature.

A thermostat needs a working source and a working sink: Venus lost its water, Mars lost its volcanism, Earth kept both.

Working the worksheet

- **Show your algebra.** The exam asks for the steps, not only the number.
- Carry **4 significant figures** through intermediate steps; round at the end.
- Use the **checkpoints** (“show that ...”) to verify your work and to keep moving if a step resists.
- Qualitative parts want **2 to 3 sentences** of physics, not an essay.
- Two parts read figures printed on the sheet (the saturation curves and the bistability diagram); argue the species and the climate branch from the curves.

Worksheet and full solutions: ips.formingworlds.space/worksheets.html.
We discuss questions, not presentations of the solutions.